SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE CODE: ITA 06 **COURSE NAME: MACHINE LEARNING**

COURSE OUTCOME COVERED

CO3: Bayes Theorem – Concept Learning – Maximum Likelihood – Minimum Description Length Principle – Bayes Optimal Classifier – Gibbs Algorithm – Naïve Bayes Classifier – Bayesian Belief Network – EM Algorithm – Probability Learning – Sample Complexity – Finite and Infinite Hypothesis Spaces – Mistake Bound Model, (BL 6)

Assessment Tool Weightage: 10%

QUESTIONS: 50 Total Marks: 25

Annexure A
Multiple Choice Questions (MCQ)

Code of Conduct:

I V.Madhulika (192312620) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

V.Madhulika

Signature of the student

1. Bayes' Theorem is a fundamental concept in probability theory. Which of the following best describes its primary purpose in updating prior beliefs when new evidence is observed in a probabilistic model?

(a) To compute only prior probabilities

(b) To update prior beliefs using new evidence

(c) To eliminate irrelevant features

(d) To maximize classification accuracy

Answer: (b) To update prior beliefs using new evidence

2. In the Naïve Bayes classifier, a strong assumption is made about the relationship between features. Which of the following statements correctly describes this assumption in practical machine learning applications?

(a) Features are completely dependent on each other

(b) Features are conditionally independent given the class

(c) Features must be continuous

(d) Features are always correlated

Answer: (b) Features are conditionally independent given the class

3. Maximum Likelihood Estimation is widely used for parameter estimation in probabilistic models. Which of the following options correctly explains the objective of MLE when fitting a model to observed data?

(a) Maximize prior probability

(b) Minimize hypothesis space

(c) Maximize likelihood of observed data

(d) Reduce computational complexity

Answer: (c) Maximize likelihood of observed data

4. The Minimum Description Length principle is often used in model selection. Which of the following best explains how MDL balances model complexity and goodness of fit in learning algorithms?

(a) Prefers highly complex models always

(b) Prefers simplest model that explains data well

(c) Ignores data noise completely

(d) Focuses only on training accuracy

Answer: (b) Prefers simplest model that explains data well

5. The Bayes Optimal Classifier is considered theoretically ideal in classification tasks. Which of the following statements best describes how it achieves the minimum possible classification error?

(a) By selecting random hypotheses

(b) By averaging predictions over all hypotheses

(c) By using only prior probabilities

(d) By minimizing dataset size

Answer: (b) By averaging predictions over all hypotheses

6. The Gibbs Algorithm is an important concept in Bayesian learning. Which of the following options best explains how it selects a hypothesis from the hypothesis space during classification?

(a) Selects hypothesis randomly from consistent set

(b) Selects hypothesis with minimum loss only

(c) Selects fixed predefined hypothesis

(d) Ignores hypothesis space

Answer: (a) Selects hypothesis randomly from consistent set

7. Naïve Bayes classifiers are often used in text classification tasks. Which of the following conditions ensures that the classifier performs effectively despite its simplifying assumptions?

(a) Strong dependence between features

(b) Large amount of labeled data

(c) Features being conditionally independent

(d) Absence of noise in dataset

Answer: (c) Features being conditionally independent

8. Bayesian Belief Networks are graphical models that represent probabilistic relationships. Which of the following best describes the role of nodes and edges in such networks?

(a) Nodes represent variables and edges represent dependencies

(b) Nodes represent datasets and edges represent features

(c) Nodes represent outputs only

(d) Edges represent noise in data

Answer: (a) Nodes represent variables and edges represent dependencies

9. The Expectation-Maximization algorithm is commonly used when dealing with incomplete data. Which of the following best explains the role of the Expectation step in this iterative algorithm?

(a) Updates model parameters

(b) Computes expected values of hidden variables

(c) Removes missing data

(d) Sorts the dataset

Answer: (b) Computes expected values of hidden variables

10. In the EM algorithm, the Maximization step updates certain parameters. Which of the following correctly describes what is updated during this step to improve model likelihood?

(a) Hidden variables only

(b) Observed data values

(c) Model parameters maximizing likelihood

(d) Training dataset size

Answer: (c) Model parameters maximizing likelihood

11. Probability learning involves estimating probability distributions from data. Which of the following best explains the goal of probability learning in the context of machine learning models?

(a) To remove all uncertainty

(b) To estimate underlying probability distributions

(c) To reduce number of features

(d) To simplify model structure

Answer: (b) To estimate underlying probability distributions

12. Sample complexity is an important concept in learning theory. Which of the following best describes what sample complexity indicates about a learning algorithm's performance?

(a) Amount of computation required

(b) Number of samples needed for learning

(c) Model accuracy only

(d) Memory consumption

Answer: (b) Number of samples needed for learning

13. A finite hypothesis space contains a limited number of possible hypotheses. Which of the following best explains how this limitation impacts the learning process and generalization ability?

(a) Improves flexibility always

(b) Restricts possible models considered

(c) Eliminates need for data

(d) Guarantees perfect accuracy

Answer: (b) Restricts possible models considered

14. In contrast to finite hypothesis spaces, infinite hypothesis spaces allow more flexibility. Which of the following best describes the characteristics of such hypothesis spaces in learning models?

(a) Limited to discrete functions

(b) Allows continuous and complex models

(c) Contains no hypotheses

(d) Simplifies computation always

Answer: (b) Allows continuous and complex models

15. The mistake bound model is used to evaluate learning algorithms. Which of the following best describes what is measured by the mistake bound in online learning scenarios?

(a) Training time

(b) Maximum number of mistakes made

(c) Memory usage

(d) Dataset size

Answer: (b) Maximum number of mistakes made

16. Prior probability is a key concept in Bayesian inference. Which of the following best describes what prior probability represents before observing any new evidence?

(a) Probability after evidence

(b) Initial belief before observing data

(c) Conditional probability only

(d) Joint probability

Answer: (b) Initial belief before observing data

17. Posterior probability is calculated after incorporating evidence. Which of the following best describes how posterior probability is derived using Bayes' Theorem?

(a) Using only likelihood

(b) Using prior and likelihood with evidence

(c) Ignoring prior information

(d) Using random sampling

Answer: (b) Using prior and likelihood with evidence

18. Likelihood plays a central role in statistical inference. Which of the following best explains what likelihood represents in relation to observed data and hypotheses?

(a) Probability of hypothesis given data

(b) Probability of data given hypothesis

(c) Probability of prior only

(d) Probability of error

Answer: (b) Probability of data given hypothesis

19. Naïve Bayes classifiers are widely used in real-world applications. Which of the following best describes the type of problems where this algorithm is most effective?

(a) Clustering problems

(b) Classification problems with high-dimensional data

(c) Regression problems

(d) Sorting problems

Answer: (b) Classification problems with high-dimensional data

20. Overfitting is a common issue in machine learning models. Which of the following best describes a situation where a model is considered to be overfitting the training data?

(a) Model performs well on unseen data

(b) Model captures noise instead of pattern

(c) Model is too simple

(d) Model ignores data

Answer: (b) Model captures noise instead of pattern

21. Underfitting occurs when a model fails to capture patterns in data. Which of the following best describes the characteristics of an underfitted model in practice?

(a) High accuracy always

(b) Fails to learn patterns from data

(c) Too complex model

(d) Overtrained model

Answer: (b) Fails to learn patterns from data

22. The EM algorithm is iterative and requires a stopping condition. Which of the following best explains when the algorithm is considered to have converged?

(a) When data changes

(b) When parameters stabilize across iterations

(c) When iterations increase

(d) When accuracy decreases

Answer: (b) When parameters stabilize across iterations

23. Bayesian learning combines prior knowledge with observed data. Which of the following best describes how these two components interact during the learning process?

(a) Prior replaces data

(b) Data replaces prior

(c) Both are combined to update belief

(d) Neither is used

Answer: (c) Both are combined to update belief

24. Concept learning involves identifying a suitable hypothesis. Which of the following best explains the main goal of concept learning in machine learning systems?

(a) To reduce computation

(b) To find hypothesis consistent with data

(c) To eliminate features

(d) To increase dataset size

Answer: (b) To find hypothesis consistent with data

25. Hypothesis space search is a key component of concept learning. Which of the following best describes how searching through hypothesis spaces helps in selecting an appropriate model?

(a) Random selection of hypothesis

(b) Exploring possible hypotheses to find best fit

(c) Ignoring data patterns

(d) Reducing dataset size

Answer: (b) Exploring possible hypotheses to find best fit

26. In Bayesian inference, normalization ensures probabilities sum to one. What role does the evidence term play in Bayes' Theorem?

(a) It acts as a scaling factor

(b) It removes prior probability

(c) It increases model complexity

(d) It selects hypothesis randomly

Answer: (a) It acts as a scaling factor

27. In Naïve Bayes classification, what is the primary reason for using logarithms when computing probabilities?

(a) To increase accuracy

(b) To avoid numerical underflow

(c) To remove dependencies

(d) To simplify dataset

Answer: (b) To avoid numerical underflow

28. Which of the following best describes conditional probability?

(a) Probability of two events occurring together

(b) Probability of an event given another event has occurred

(c) Probability of independent events

(d) Probability of random variables

Answer: (b) Probability of an event given another event has occurred

29. Which factor most affects the performance of a Naïve Bayes classifier in real-world datasets?

(a) Number of classes

(b) Independence assumption violation

(c) Dataset size only

(d) Output format

Answer: (b) Independence assumption violation

30. In Bayesian networks, what ensures that the model remains computationally efficient?

(a) Large dataset

(b) Conditional independence

(c) Continuous variables

(d) High dimensionality

Answer: (b) Conditional independence

31. Which of the following best describes the role of prior knowledge in Bayesian learning?

(a) Eliminates need for data

(b) Guides learning before observing data

(c) Reduces accuracy

(d) Replaces hypothesis

Answer: (b) Guides learning before observing data

32. In Maximum Likelihood Estimation, what happens when the dataset size increases significantly?

(a) Prior dominates

(b) Likelihood estimation becomes more accurate

(c) Model becomes random

(d) Complexity decreases

Answer: (b) Likelihood estimation becomes more accurate

33. Which of the following is a limitation of MLE?

(a) Uses too much prior knowledge

(b) Ignores prior information
(c) Cannot handle data
(d) Always overfits

Answer: (b) Ignores prior information

34. Which of the following best describes Bayesian estimation compared to MLE?
(a) Uses only likelihood
(b) Uses both prior and likelihood
(c) Ignores uncertainty
(d) Works only for small datasets

Answer: (b) Uses both prior and likelihood

35. What is the main advantage of Laplace smoothing in Naïve Bayes?
(a) Removes features
(b) Handles zero probability problem
(c) Reduces dataset size
(d) Increases bias only

Answer: (b) Handles zero probability problem

36. Which of the following best defines a hypothesis in machine learning?
(a) Final output
(b) Assumed model for mapping inputs to outputs
(c) Dataset
(d) Feature vector

Answer: (b) Assumed model for mapping inputs to outputs

37. What is the purpose of hypothesis space in learning algorithms?
(a) Store data
(b) Define all possible models
(c) Reduce training time
(d) Increase features

Answer: (b) Define all possible models

38. Which of the following best explains generalization in machine learning?
(a) Memorizing training data
(b) Performing well on unseen data
(c) Reducing dataset size
(d) Increasing parameters

Answer: (b) Performing well on unseen data

39. What is the effect of noisy data on learning algorithms?
(a) Improves accuracy

(b) Leads to overfitting
(c) Eliminates features
(d) Reduces complexity

Answer: (b) Leads to overfitting

40. In EM algorithm, what is the main challenge during initialization?
(a) Dataset size
(b) Choosing initial parameters
(c) Output format
(d) Removing features

Answer: (b) Choosing initial parameters

41. Which of the following best describes convergence in iterative algorithms?
(a) Increasing error
(b) Stabilization of parameters
(c) Random updates
(d) Dataset reduction

Answer: (b) Stabilization of parameters

42. Which of the following is a characteristic of probabilistic models?
(a) Deterministic outputs
(b) Handling uncertainty
(c) Ignoring data
(d) Fixed predictions

Answer: (b) Handling uncertainty

43. What is the role of likelihood in Bayesian inference?
(a) Represents prior
(b) Represents evidence
(c) Measures fit of data to hypothesis
(d) Removes uncertainty

Answer: (c) Measures fit of data to hypothesis

44. Which of the following best describes overfitting?
(a) Model too simple
(b) Model fits noise in training data
(c) Model ignores data
(d) Model has low variance

Answer: (b) Model fits noise in training data

45. Which of the following best describes underfitting?
(a) Model too complex

(b) Model fails to capture patterns
(c) Model memorizes data
(d) Model has high variance

Answer: (b) Model fails to capture patterns

46. What is the role of training data in learning algorithms?
(a) Defines output only
(b) Guides model learning
(c) Reduces complexity
(d) Eliminates hypothesis

Answer: (b) Guides model learning

47. Which factor improves model generalization?
(a) Increasing noise
(b) Proper regularization
(c) Reducing data
(d) Ignoring validation

Answer: (b) Proper regularization

48. Which of the following best explains feature selection?
(a) Removing dataset
(b) Selecting relevant features
(c) Increasing features
(d) Ignoring attributes

Answer: (b) Selecting relevant features

49. Which of the following best describes classification problems?
(a) Predict continuous values
(b) Assign labels to data
(c) Cluster data
(d) Reduce dimensions

Answer: (b) Assign labels to data

50. Which of the following best describes regression problems?
(a) Predict discrete labels
(b) Predict continuous values
(c) Cluster data
(d) Classify data

Answer: (b) Predict continuous values

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	20	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	15	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand